

ZHIGANSK, Russian Federation

WMO: 243430

Lat: **66.767N**Lon: **123.400E**Elev: **92**StdP: **100.22**Time Zone: **9.00 (E09)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-48.3	-46.8	-50.2	0.0	-48.3	-48.6	0.0	-46.7	7.5	-26.7	6.5	-29.1	1.2	140	0.784

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.1	26.7	17.5	24.7	16.9	22.6	15.9	19.2	23.8	18.0	22.7	16.8	21.7	2.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.5	12.7	21.6	16.1	11.6	20.4	14.7	10.6	19.2	55.4	23.6	51.5	22.7	47.8	21.8	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.5	6.6	5.8	-50.1	30.0	2.6	1.9	-52.0	31.4	-53.5	32.5	-54.9	33.5	-56.8	34.9		
			-50.4	21.2	2.7	1.4	-52.4	22.2	-53.9	23.0	-55.4	23.8	-57.4	24.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-10.5	-36.9	-33.8	-21.7	-8.5	2.5	13.2	16.2	12.2	3.8	-9.7	-27.7	-37.0	(6)
(7)		DBStd	20.31	6.21	6.33	7.59	6.17	5.19	4.86	4.20	3.75	4.23	7.41	7.11	6.50	(7)
(8)		HDD10.0	7893	1454	1228	984	554	238	23	3	20	191	609	1132	1456	(8)
(9)		HDD18.3	10564	1712	1461	1242	804	492	164	94	193	436	868	1382	1714	(9)
(10)		CDD10.0	409	0	0	0	0	4	119	194	88	4	0	0	0	(10)
(11)		CDD18.3	39	0	0	0	0	0	9	27	3	0	0	0	0	(11)
(12)		CDH23.3	323	0	0	0	0	0	93	212	18	0	0	0	0	(12)
(13)		CDH26.7	54	0	0	0	0	0	16	37	2	0	0	0	0	(13)
(14)	Wind	WSAvg	2.6	2.3	2.2	2.5	3.0	3.3	3.0	2.9	2.7	2.6	2.6	2.1	2.1	(14)
(15)	Precipitation	PrecAvg	274	9	8	11	13	19	32	48	38	36	31	17	13	(15)
(16)		PrecMax	364	16	22	31	29	56	85	161	86	59	47	43	29	(16)
(17)		PrecMin	182	3	2	2	1	0	6	0	10	14	13	5	4	(17)
(18)		PrecStd	53	4	4	7	7	14	20	38	21	13	8	9	7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-19.2	-14.8	-0.7	8.4	20.3	28.8	29.7	25.3	19.1	3.2	-8.1	-18.3	(19)
(20)			MCWB	-19.5	-15.1	-2.9	3.1	10.8	17.6	18.6	17.7	11.8	2.0	-8.7	-18.5	(20)
(21)		2%	DB	-22.3	-18.7	-4.5	5.1	15.5	25.9	27.6	22.6	15.4	1.5	-11.3	-22.4	(21)
(22)			MCWB	-22.5	-19.0	-6.0	0.9	8.0	16.1	18.2	16.5	10.0	0.5	-11.6	-22.7	(22)
(23)		5%	DB	-25.1	-21.6	-7.4	3.0	12.9	23.7	25.8	20.9	12.4	0.2	-14.1	-24.6	(23)
(24)			MCWB	-25.4	-21.9	-8.5	-0.1	6.7	15.2	17.7	15.4	8.7	-0.5	-14.5	-24.9	(24)
(25)		10%	DB	-27.9	-24.4	-10.2	1.2	10.5	21.7	23.9	18.9	10.2	-0.9	-17.6	-27.2	(25)
(26)			MCWB	-28.1	-24.6	-11.1	-1.2	5.3	14.2	17.2	14.4	7.4	-1.8	-17.9	-27.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-19.5	-15.1	-2.7	3.4	10.8	19.2	21.3	19.8	12.9	2.5	-8.7	-18.6	(27)
(28)			MCDWB	-19.2	-14.8	-0.9	7.6	19.6	25.9	26.6	24.2	17.3	2.9	-8.2	-18.3	(28)
(29)		2%	WB	-22.5	-19.0	-5.9	1.6	8.6	17.6	19.9	17.7	11.0	0.7	-11.6	-22.6	(29)
(30)			MCDWB	-22.3	-18.8	-4.6	4.3	14.2	23.5	24.4	20.8	13.3	1.1	-11.4	-22.5	(30)
(31)		5%	WB	-25.3	-21.9	-8.4	0.2	7.1	16.4	18.8	16.3	9.2	-0.3	-14.4	-24.8	(31)
(32)			MCDWB	-25.1	-21.7	-7.4	2.7	12.1	22.3	23.4	19.8	12.0	0.2	-14.1	-24.7	(32)
(33)		10%	WB	-28.1	-24.6	-11.2	-1.3	5.7	15.0	17.8	15.0	7.6	-1.7	-17.8	-27.3	(33)
(34)			MCDWB	-27.9	-24.4	-10.3	1.3	10.2	20.3	22.6	18.1	9.9	-1.0	-17.6	-27.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.5	6.7	9.1	10.3	7.9	9.4	9.1	8.7	7.1	5.4	5.9	5.1	(35)
(36)			MCDBR	7.9	7.6	9.6	10.1	10.8	12.9	12.3	11.7	11.5	4.4	7.0	6.9	(36)
(37)			MCWBR	7.8	7.5	8.6	7.2	6.1	6.2	5.6	6.5	6.6	4.0	6.9	6.9	(37)
(38)		5% WB	MCDBR	7.8	7.6	9.6	9.1	10.0	11.6	9.7	9.8	10.2	4.4	7.0	6.8	(38)
(39)			MCWBR	7.7	7.5	8.4	6.8	5.8	6.0	5.3	6.2	6.5	4.2	6.9	6.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.186	0.235	0.286	0.326	0.358	0.386	0.477	0.456	0.317	0.250	0.181	N/A	(40)	
(41)		taud	1.871	2.049	2.073	2.081	2.263	2.348	2.044	2.050	2.465	2.196	1.893	N/A	(41)	
(42)		Ebn at Noon	188	604	753	818	835	819	709	673	746	584	177	0	(42)	
(43)		Edh at Noon	15	54	91	119	112	106	140	125	65	45	13	0	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.07	0.78	2.59	4.86	5.69	5.78	5.30	3.61	2.11	0.70	0.18	0.00	(44)	
(45)		RadStd	0.01	0.04	0.11	0.20	0.42	0.50	0.41	0.44	0.21	0.11	0.01	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page